

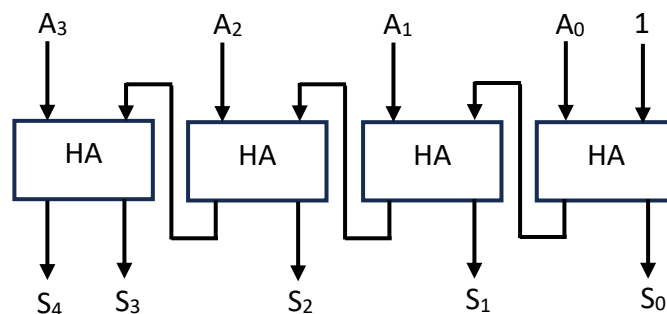


Wayamba University of Sri Lanka
Diploma in Information Technology – Intake 15
DIT11022 - Computer Systems and Architectures
Assignment 01

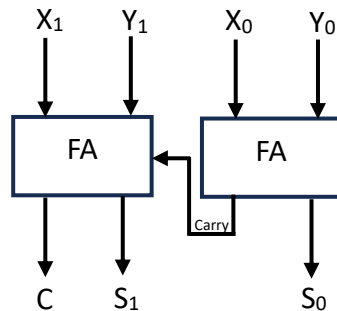
ANSWER ALL QUESTIONS

Due Date: 31/ 08/ 2026

1. Design a combinational logic circuit with four inputs (A, B, C, D) and two outputs (Y1, Y2). Y1 outputs high if the input has an odd number of ones and Y2 outputs high if the binary input value is a prime number.
 - a. Construct the complete truth table listing all 16 input combinations
 - b. Derive the simplified Boolean expression for each of the outputs using Karnaugh maps (K-maps).
 - c. Draw the final logic gate diagram using standard universal gates based on your simplified expressions.
2. Design a combinational logic circuit that accepts a 4-bit BCD input (A, B, C, D), where A is MSB and D is LSB representing decimal digits 0 through 9. The circuit must generate a 4-bit Excess-3 output (W, X, Y, Z), where W is MSB and Z is LSB by adding 3 (0011) to the BCD input.
 - a. Construct the complete truth table listing all 16 input combinations
 - b. Derive the simplified Boolean expression for each of the outputs using Karnaugh maps (K-maps).
 - c. Draw the final logic gate diagram using basic gates (AND, OR, NOT) based on your simplified expressions.
3. Design a combinational circuit that adds one to a 4-bit binary number. For example, if the input of the circuit is 1101, the output is 1110. Provide a truth table of the combinational circuit. Get the simplified Sum of Product expression for each output using k-map and draw the circuit.
4. Design the circuit for the above problem using four half-adders.



5. A combinational circuit produces the binary sum of two 2-bit numbers, X_1, X_0 and Y_1, Y_0 . The outputs are C, S_1 , and S_0 . Provide a truth table of the combinational circuit. Get the simplified Sum of Product expression for each output using k-map and draw the circuit. X_1 and Y_1 are the MSB of input numbers
 X_0 and Y_0 are the LSB of input numbers
 C, S_1, S_0 are Carry Bit, MSB, LSB of the output
6. Design the circuit for the above problem using two full adders.



7. Design a combinational logic circuit with three inputs (A, B, C) and two outputs (Y_1, Y_2). Output Y_1 is HIGH if the binary input is an even number (including zero), and output Y_2 is HIGH if the input value is greater than 3. Derive the truth table, find simplified Boolean expressions using K-maps, and draw the final gate diagram.